

Proposed Solution to #5820 SSMJ

Solution proposed by Prakash Pant, The University of Vermont, Bardiya, Nepal.

Problem proposed by Michel Bataille, Rouen, France

Statement of the Problem:

Let r, s be integers with $r \geq s \geq 0$ and let the real number x satisfy $|x| < 1$. Show that

$$\sum_{n=0}^{\infty} \binom{r+n}{s} x^n = \sum_{j=0}^s \binom{r}{s-j} \frac{x^j}{(1-x)^{j+1}}$$

Solution of the Problem:

We encode the binomial coefficient as follows:

$$\binom{r+n}{s} = [t^s](1+t)^{r+n}$$

where $[t^s]$ means the coefficient of t^s . Then,

$$\text{L.H.S} = \sum_{n=0}^{\infty} [t^s](1+t)^{r+n} x^n$$

Since sum of coefficients is coefficient of sum,

$$[t^s](1+t)^r \sum_{n=0}^{\infty} (x+xt)^n$$

Since $|x| < 1$ and t is an arbitrary variable which can be chosen to be small enough such that $|x||1+t| < 1$. Thus, using infinite geometric series,

$$[t^s](1+t)^r \frac{1}{1-x-xt}$$
$$[t^s](1+t)^r \left(\frac{1}{1-x} \right) \left(\frac{1}{1-\frac{x}{1-x}t} \right)$$

Again if t is small enough and since $|x| < 1$, $\left|\frac{xt}{1-x}\right| < 1$. Thus,

$$[t^s](1+t)^r \left(\frac{1}{1-x}\right) \sum_{j=0}^{\infty} \frac{x^j}{(1-x)^j} t^j$$

$$[t^s] \sum_{j=0}^{\infty} \frac{x^j}{(1-x)^{j+1}} (1+t)^r t^j$$

Again, since coefficient of sum is sum of coefficients,

$$\sum_{j=0}^{\infty} \frac{x^j}{(1-x)^{j+1}} [t^s](1+t)^r t^j = \sum_{j=0}^{\infty} \frac{x^j}{(1-x)^{j+1}} [t^{s-j}](1+t)^r$$

$$\sum_{j=0}^{\infty} \frac{x^j}{(1-x)^{j+1}} \binom{r}{s-j}$$

Since $s-j$ needs to be positive for non zero binomial coefficient, $s-j \geq 0 \implies s \geq j \implies j \leq s$. Thus,

$$\sum_{j=0}^s \binom{r}{s-j} \frac{x^j}{(1-x)^{j+1}} = \text{R.H.S}$$

which concludes the proof.